

PowerShell Introduction Lab

Mapping classic Windows CMD commands to their PowerShell equivalents

Project Summary

This lab demonstrates basic Windows command-line usage and the matching PowerShell cmdlets for file system navigation, file creation, copying, deletion, and output display. Each step below shows a short instruction, the exact command to run, and a screenshot of the expected result.

Before You Start

- Windows 10 or 11. No administrator rights are needed.
- Open Command Prompt for Part 1, and PowerShell for Parts 2 and 3.
- Replace <YourUsername> in every path with your own Windows account name. To find it, run the command below.

```
# Show your username and home folder path
echo %USERNAME%
echo %USERPROFILE%
```

Create the two sample files this lab uses. Run this first, or the copy and delete steps will fail.

```
# Setup - create the sample files
cd C:\Users\<YourUsername>\Documents
echo Hello from PowerShell > file1.txt
echo Hello from PowerShell > file2.txt
```

Command Mapping

A quick reference for every command used in this lab.

Task	CMD Command	PowerShell Equivalent
List files and folders	dir	Get-ChildItem
Change directory	cd	Set-Location
Copy a file	copy	Copy-Item
Delete a file	del	Remove-Item
Print a message	echo	Write-Host / Write-Output
Create a folder	mkdir	New-Item -ItemType Directory
Create an empty file	type nul > file	New-Item -ItemType File
Read a file	type	Get-Content

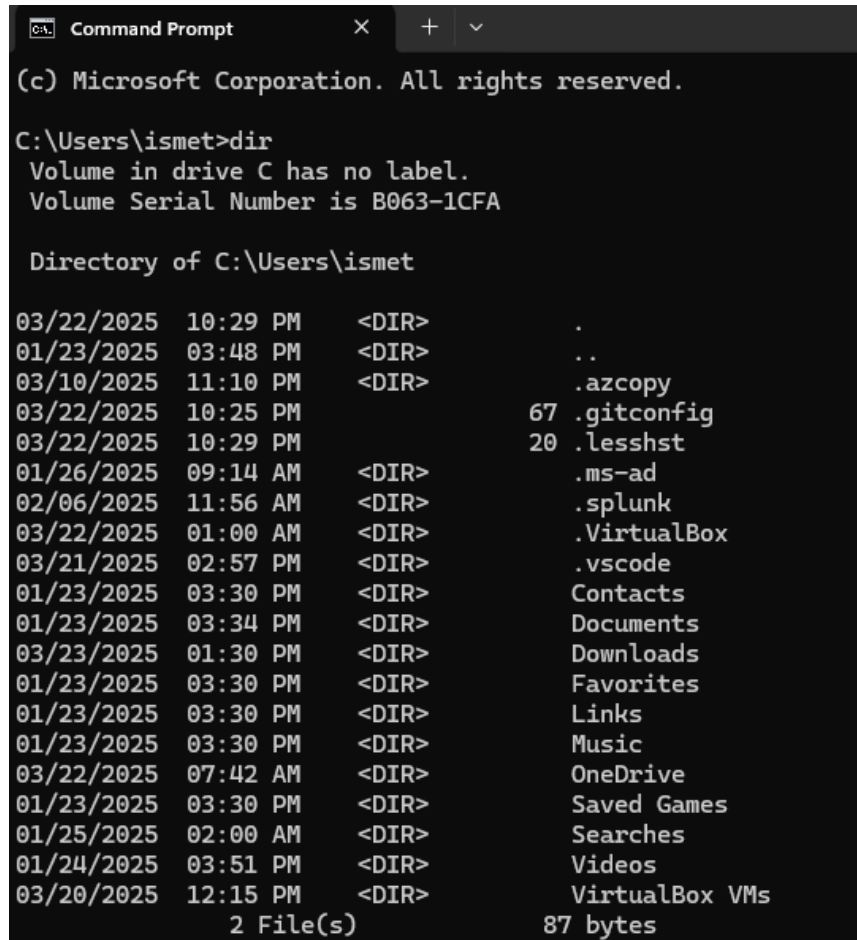
Part 1 - CMD Commands

Open Command Prompt (press Windows key, type cmd, press Enter) and run each command in order.

Step 1 - dir

List every file and folder in the current directory.

```
# List the contents of the current folder
dir
```



```
(c) Microsoft Corporation. All rights reserved.

C:\Users\ismet>dir
Volume in drive C has no label.
Volume Serial Number is B063-1CFA

Directory of C:\Users\ismet

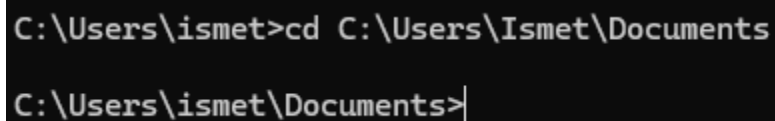
03/22/2025  10:29 PM  <DIR>          .
01/23/2025  03:48 PM  <DIR>          ..
03/10/2025  11:10 PM  <DIR>          .azcopy
03/22/2025  10:25 PM             67 .gitconfig
03/22/2025  10:29 PM             20 .lessht
01/26/2025  09:14 AM  <DIR>          .ms-ad
02/06/2025  11:56 AM  <DIR>          .splunk
03/22/2025  01:00 AM  <DIR>          .VirtualBox
03/21/2025  02:57 PM  <DIR>          .vscode
01/23/2025  03:30 PM  <DIR>          Contacts
01/23/2025  03:34 PM  <DIR>          Documents
03/23/2025  01:30 PM  <DIR>          Downloads
01/23/2025  03:30 PM  <DIR>          Favorites
01/23/2025  03:30 PM  <DIR>          Links
01/23/2025  03:30 PM  <DIR>          Music
03/22/2025  07:42 AM  <DIR>          OneDrive
01/23/2025  03:30 PM  <DIR>          Saved Games
01/25/2025  02:00 AM  <DIR>          Searches
01/24/2025  03:51 PM  <DIR>          Videos
03/20/2025  12:15 PM  <DIR>          VirtualBox VMs
                2 File(s)                87 bytes
```

dir output showing the contents of the user folder

Step 2 - cd

Move into the Documents folder. This becomes your working directory for the rest of Part 1.

```
# Change directory
cd C:\Users\<YourUsername>\Documents
```



```
C:\Users\ismet>cd C:\Users\Ismet\Documents

C:\Users\ismet\Documents>|
```

The prompt changes to show the new location

Step 3 - copy

Make a duplicate of file2.txt and name the copy file3.txt.

```
# Copy a file to a new name
copy "C:\Users\<YourUsername>\Documents\file2.txt" "C:\Users\<YourUsername>\Documents\
file3.txt"

# Confirm the copy exists
dir
```

Note: Always wrap a path in double quotes if it contains a space.

```
C:\Users\ismet\Documents>copy "C:\Users\ismet\Documents\file2.txt" "C:\Users
\ismet\Documents\file3.txt"
        1 file(s) copied.

C:\Users\ismet\Documents>dir
Volume in drive C has no label.
Volume Serial Number is B063-1CFA

Directory of C:\Users\ismet\Documents

03/23/2025  02:04 PM    <DIR>          .
03/22/2025  10:29 PM    <DIR>          ..
03/23/2025  01:50 PM                0 copy
03/23/2025  01:57 PM             16 file1.txt
03/23/2025  02:01 PM             16 file2.txt
03/23/2025  02:01 PM             16 file3.txt
                4 File(s)              48 bytes
                2 Dir(s) 395,459,633,152 bytes free

C:\Users\ismet\Documents>
```

One file copied, then dir confirms file3.txt exists

Step 4 - del

Delete file2.txt. Use the full path so the command works from any directory.

```
# Delete a file
del C:\Users\<YourUsername>\Documents\file2.txt
```

Note: In the screenshot the first attempt fails because the prompt had moved up a level with `cd ..` and the relative path no longer resolved. Giving the full path fixes it.

```
C:\Users\ismet\Documents>cd ..

C:\Users\ismet>del C:\Users\ismet\Documents\file2.txt
The system cannot find the path specified.

C:\Users\ismet>del C:\Users\ismet\Documents\file2.txt

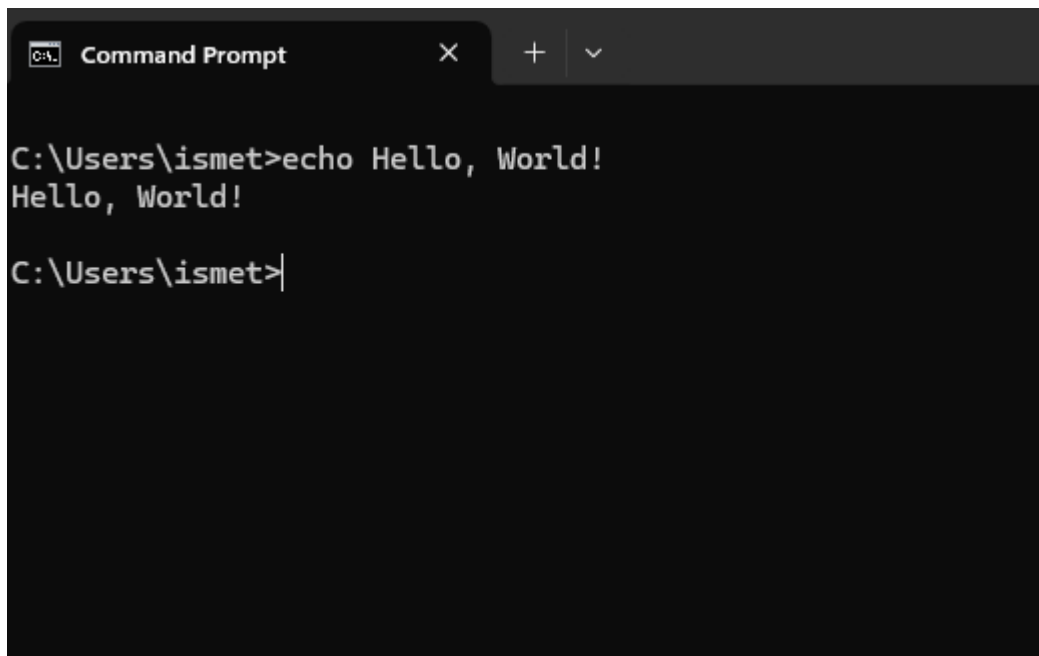
C:\Users\ismet>|
```

The first attempt fails, the second succeeds

Step 5 - echo

Print a message to the screen.

```
# Print text to the console
echo Hello, World!
```



```
Command Prompt

C:\Users\ismet>echo Hello, World!
Hello, World!

C:\Users\ismet>|
```

echo prints the message back

Step 6 - mkdir and type nul

Create a new folder, then create an empty file inside it.

```
# Create a folder
mkdir "C:\Users\<YourUsername>\Documents\CMD Scripts"

# Create an empty file inside it
type nul > "C:\Users\<YourUsername>\Documents\CMD Scripts\test.txt"

# Verify
dir "C:\Users\<YourUsername>\Documents\CMD Scripts"
```

Note: `type nul > filename` is the CMD trick for creating a zero-byte file. There is no dedicated command for it.

Part 2 - PowerShell Equivalents

Close Command Prompt and open PowerShell (press Windows key, type powershell, press Enter). Run the same tasks using cmdlets.

Step 7 - Get-ChildItem

List the contents of a folder. This is the PowerShell replacement for dir.

```
# List files and folders
Get-ChildItem

# dir and ls are built-in aliases for the same cmdlet
dir
ls
```

Note: Get-ChildItem returns objects, not plain text. That is why you can sort and filter the results with the pipeline.

```
PowerShell 7.6.0-preview.3
PS C:\Users\ismet> Get-ChildItem

Directory: C:\Users\ismet

Mode                LastWriteTime         Length Name
----                -
d-----          3/10/2025 11:10 PM             .azcopy
d-----          1/26/2025  8:14 AM             .ms-ad
d-----           2/6/2025 10:56 AM             .splunk
d-----          3/22/2025  1:00 AM      .VirtualBox
d-----          3/21/2025  2:57 PM             .vscode
d-r--          1/23/2025  2:30 PM           Contacts
d-----          1/23/2025  2:34 PM           Documents
d-r--          3/23/2025  1:30 PM           Downloads
d-r--          1/23/2025  2:30 PM           Favorites
d-r--          1/23/2025  2:30 PM           Links
d-r--          1/23/2025  2:30 PM           Music
lar--          3/22/2025  7:42 AM           OneDrive
d-r--          1/23/2025  2:30 PM       Saved Games
d-r--          1/25/2025  1:00 AM           Searches
d-r--          1/24/2025  2:51 PM           Videos
d-----          3/20/2025 12:15 PM VirtualBox VMs
-a---          3/22/2025 10:25 PM           67 .gitconfig
-a---          3/22/2025 10:29 PM           20 .lessht

PS C:\Users\ismet> |
```

Get-ChildItem returns objects with Mode, LastWriteTime, Length and Name

Step 8 - Set-Location

Change the working directory. This is the PowerShell replacement for cd.

```
# Change directory
Set-Location C:\Users\<YourUsername>\Documents

# Confirm where you are
Get-Location
```

```
PS C:\Users\ismet> Set-Location C:\Users\Ismet\Documents
PS C:\Users\ismet\Documents> |
```

The prompt updates to the new path

Step 9 - Out-File and Get-Content

Write text into a new file, then read it back.

```
# Write text to a file
"Hello-World" | Out-File -FilePath C:\Users\<YourUsername>\Documents\file10.txt

# Read the file back
Get-Content C:\Users\<YourUsername>\Documents\file10.txt
```

Note: *type and cat are both aliases for Get-Content, so all three commands do the same thing.*

```
PS C:\Users\ismet> Set-Location C:\Users\Ismet\Documents
PS C:\Users\ismet\Documents> "Hello-World" | Out-file -FilePath C:\Users\ismet\Documents\file10.txt
PS C:\Users\ismet\Documents> type C:\Users\ismet\Documents\file10.txt
Hello-World
PS C:\Users\ismet\Documents> |
```

The file is written, then read back with type (an alias for Get-Content)

Step 10 - Remove-Item

Delete the file you just created, then confirm it is gone.

```
# Delete a file
Remove-Item C:\Users\<YourUsername>\Documents\file10.txt

# Try to read it - an error here proves the delete worked
Get-Content C:\Users\<YourUsername>\Documents\file10.txt
```

Note: *To delete a folder and everything inside it, add -Recurse. Add -Force to remove hidden or read-only items.*

```
PS C:\Users\ismet\Documents> Remove-Item C:\Users\ismet\Documents\file10.txt
PS C:\Users\ismet\Documents> type C:\Users\ismet\Documents\file10.txt
Get-Content: Cannot find path 'C:\Users\ismet\Documents\file10.txt' because it does not exist.
PS C:\Users\ismet\Documents> |
```

The "cannot find path" error confirms the file was deleted

Step 11 - Write-Host and Write-Output

Print a message to the screen.

```
# Print straight to the console
Write-Host "Hello World!"

# Send the value down the pipeline so it can be captured
Write-Output "Hello World!"
```

Note: *Use Write-Output when the value needs to be reused. Use Write-Host only for screen-only messages such as coloured status text.*

```
PS C:\Users\ismet\Documents> Write-Host "Hello World!"
Hello World!
PS C:\Users\ismet\Documents> |
```

Write-Host prints the message

Part 3 - Create a Folder and File

Step 12 - New-Item

Create a folder named "PowerShell Scripts", move into it, and create an empty test.txt inside.

```
# Go to Documents
Set-Location C:\Users\<YourUsername>\Documents

# Create the folder
New-Item -Name "PowerShell Scripts" -ItemType Directory

# Move into it - quotes are required, the name has a space
Set-Location "C:\Users\<YourUsername>\Documents\PowerShell Scripts"

# Create an empty file
New-Item -Name "test.txt" -ItemType File

# Verify
Get-ChildItem
```

Note: If you run `cd PowerShell Scripts` without quotes, PowerShell reads "Scripts" as a second argument and returns "A positional parameter cannot be found that accepts argument 'Scripts'". Quoting the full path fixes it. This error is visible in the screenshot below.

```
PS C:\Users\ismet\Documents> New-Item -Path "C:\Users\ismet\Documents" -Name "PowerShell Scripts" -ItemType "Directory"

Directory: C:\Users\ismet\Documents

Mode                LastWriteTime         Length Name
----                -
d-----          3/23/2025  2:19 PM                PowerShell Scripts

PS C:\Users\ismet\Documents> cd PowerShell Scripts
Set-Location: A positional parameter cannot be found that accepts argument 'Scripts'.
PS C:\Users\ismet\Documents> cd "PowerShell Scripts"
Set-Location: Cannot find path 'C:\Users\ismet\Documents\PowerShell Scripts' because it does not exist.
PS C:\Users\ismet\Documents> cd 'PowerShell Scripts'
Set-Location: Cannot find path 'C:\Users\ismet\Documents\PowerShell Scripts' because it does not exist.
PS C:\Users\ismet\Documents> cd C:\Users\ismet\Documents\PowerShell Scripts
PS C:\Users\ismet\Documents\PowerShell Scripts> ls
PS C:\Users\ismet\Documents\PowerShell Scripts> New-Item -Name "test.txt" -ItemType "File"

Directory: C:\Users\ismet\Documents\PowerShell Scripts

Mode                LastWriteTime         Length Name
----                -
-a-----          3/23/2025  2:22 PM              0 test.txt

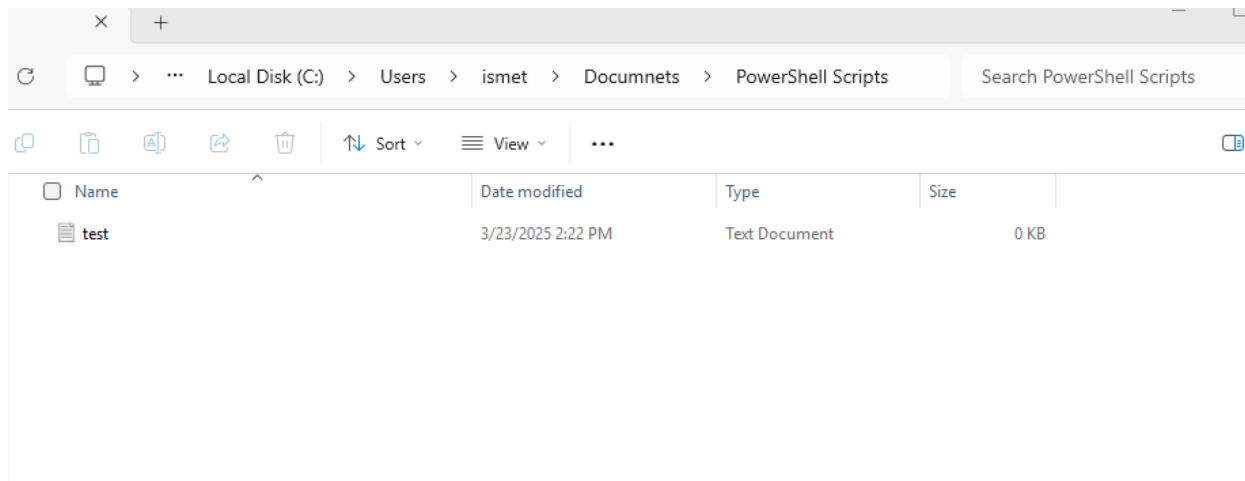
PS C:\Users\ismet\Documents\PowerShell Scripts> |
```

The quoting errors, then the successful folder and file creation

Step 13 - Verify in File Explorer

Open the folder in File Explorer to confirm the work outside the terminal.

```
# Open the folder in File Explorer
Invoke-Item "C:\Users\<YourUsername>\Documents\PowerShell Scripts"
```



test.txt visible in File Explorer, 0 KB as expected

Step 14 - Cleanup

Remove everything the lab created so you can run it again from a clean state.

```
# Move out of the folder first, or the delete will fail
Set-Location C:\Users\<YourUsername>\Documents

# Delete the lab files
Remove-Item file1.txt, file2.txt, file3.txt, file10.txt -ErrorAction SilentlyContinue

# Delete the lab folders and everything inside them
Remove-Item "PowerShell Scripts" -Recurse -Force -ErrorAction SilentlyContinue
Remove-Item "CMD Scripts" -Recurse -Force -ErrorAction SilentlyContinue
```

Useful Commands for Going Further

These four cmdlets let you discover the rest of PowerShell without leaving the terminal.

```
# See worked examples for any cmdlet
Get-Help Get-ChildItem -Examples

# Find every cmdlet that acts on items
Get-Command -Noun Item

# Discover which cmdlet a familiar command maps to
Get-Alias dir

# Inspect the properties available on an object
Get-ChildItem | Get-Member
```

Learning Outcome

After completing this lab, I practiced mapping CMD commands to PowerShell cmdlets, navigating the Windows file system, creating folders and files, handling paths that contain spaces, and verifying work in both the terminal and File Explorer.